INTERNATIONAL MARITIME ORGANIZATION

E

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Agenda item 3

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ROUTEING MEASURES AND SHIP REPORTING SYSTEMS

Amendments to the traffic separation scheme "Off Delaware Bay"

Submitted by the United States

SUMMARY

Executive summary:	This document contains a proposal for an amendment to the IMO-adopted traffic separation scheme *Off Delaware Bay*. This proposed amendment is intended to maintain traditional shipping routes and continue to ensure that these navigation lanes remain free of fixed structures.
Strategic direction, if applicable:	7
Output:	7.22
Action to be taken:	Paragraph 36
Related documents:	*Ships' Routeing*, 2019 edition; SOLAS regulation V/10; resolution A.572(14), as amended; MSC.1/Circ.1060, as amended; COLREG.2/Circ.42, as corrected; NCSR 13/3/5, NCSR 13/3/6, NCSR 13/3/7 and NCSR 13/INF.6

Introduction

1 The United States proposes to amend the traffic separation scheme (TSS) *Off Delaware Bay* originally adopted by IMO on 28 October 1969. A change to the south-eastern approach lanes was implemented on 15 March 1976, and the TSS was last amended in 2000. The current scheme consists of an eastern approach, a south-eastern approach, a two-way route for use by tug and tow traffic, and a precautionary area configured to exclude shoal areas too shallow for deep draught vessels. Its arrangement separates large inbound vessels from tug and barge traffic on traditional New Jersey coastal routes. Geographical coordinates and a chartlet depicting the TSS are provided in International Maritime Organization (IMO) *Ships' Routeing*, 2019 edition.

Summary

2 This document proposes to extend both the eastern and south-eastern approaches farther offshore, extend the existing precautionary area, and establish three additional offshore precautionary areas at the convergence of shipping traffic with the TSS. The proposed precautionary area at the terminus of the eastern approach would be an approximately 79

square nautical mile area. The new precautionary area in the middle of the eastern approach would be an approximately 29 square nautical mile area. The new precautionary area at the terminus of the south-eastern approach would be an approximately 314 square nautical mile area. Finally, this proposal would extend the existing near-shore precautionary area 4.5 nautical miles (NM thereafter) offshore and would gradually widen to 11 NM. Figure 1 provides a side-by-side comparison of the current TSS and with the proposed amendments.

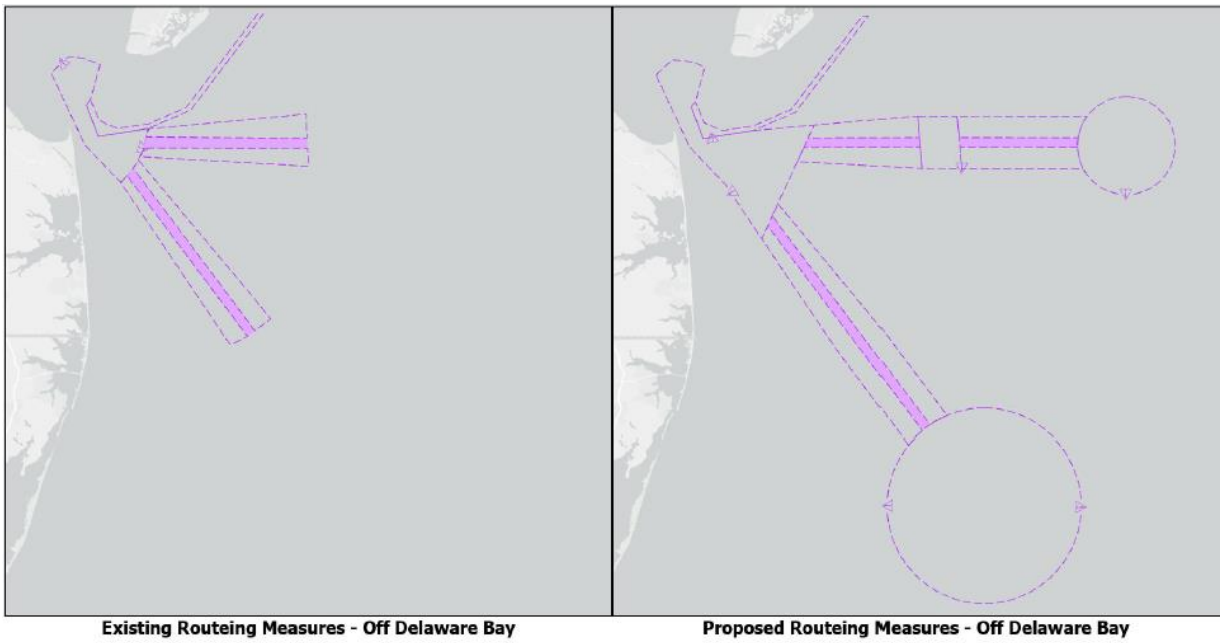

Figure 1: Proposed amendments to the TSS Off Delaware Bay

3 The United States has placed particular attention on expanded oil, natural gas, hydrokinetic, solar, nuclear, aquaculture and other non-mineral energy resources on the outer continental shelf (OCS). In identifying viable locations in the United States exclusive economic zone (EEZ) for energy production, the United States has also identified safe access routes, or "fairways," for the movement of vessel traffic proceeding to or from the United States ports, in which no artificial island or fixed structure, whether temporary or permanent, will be permitted. As such, the designation of a fairway recognizes the paramount right of navigation over all other uses in the designated areas. Based on the historical shipping trends provided in paragraph 13, coupled with the planned construction of offshore structures, the United States expects shipping transits in the approaches to Delaware Bay to consolidate into these designated fairways. Traditional shipping routes discussed in paragraph 13 and seen in figures 6 and 7 will converge perpendicular to the TSS *Off Delaware Bay* at the proposed locations. By amending the existing precautionary area, establishing three new precautionary areas, and extending the traffic lanes farther offshore, this proposal addresses the meeting of vessel traffic at critical intersections, requiring mariners to navigate with particular caution in the designated areas. A precautionary area is the preferred routeing measure as it provides a charted identifiable area where ships must navigate with particular caution. A direction of traffic flow was considered, but determined to be unnecessary and inconsistent with all other precautionary areas throughout the United States. This proposal is preventative in nature and will help provide predictable navigation corridors to accommodate shipping routes along with the evolving offshore energy infrastructure, thereby improving operational safety in increasingly congested areas of Delaware Bay. This system will continue to apply to all ships transiting the area. All other existing schemes remain unchanged. Further discussion on traffic considerations is provided in paragraphs 13 through 15. Figure 2 provides an overview of the proposal with the United States fairways depicted in green and offshore energy construction sites in blue or black.

4 In addition, this document presents all geographical coordinates in IMO *Ships' Routeing* 2019 edition description of the TSS *Off Delaware Bay* converted from the existing North American 1983 geodetic datum to the World Geodetic System of 1984 datum. Updated coordinates that have changed with respect to the shift in datum are provided in the annex.

5 A study, Port Access Route Study: Off Delaware Bay, is referenced in document NCSR 13/INF.6. The study aimed to characterize whether amendments to existing, or additional routeing measures are necessary to improve navigation safety owing to factors such as planned or potential offshore energy development, current port capabilities and planned improvements, increased vessel traffic, existing and potential anchorage areas, changing vessel traffic patterns, meteorological and oceanographic effects, or navigational difficulty in the proposal area. An in-depth analysis of the history of groundings, collisions, or damage to the marine environment can be found in section E of the study. There has been no significant growth in these incidents that is precipitating the establishment of these precautionary areas. This proposal is submitted in advance of offshore construction as a preventative measure to ensure the safety of navigation in the area.

Description of the area

6 The location of the existing TSS and two-way route, proposed precautionary areas, TSS amendments, the United States. fairways, and offshore energy construction sites are provided in figure 2. Geographic positions, number, edition and geodetic datum of the reference chart used to define the amended routeing system, are set out in the annex.

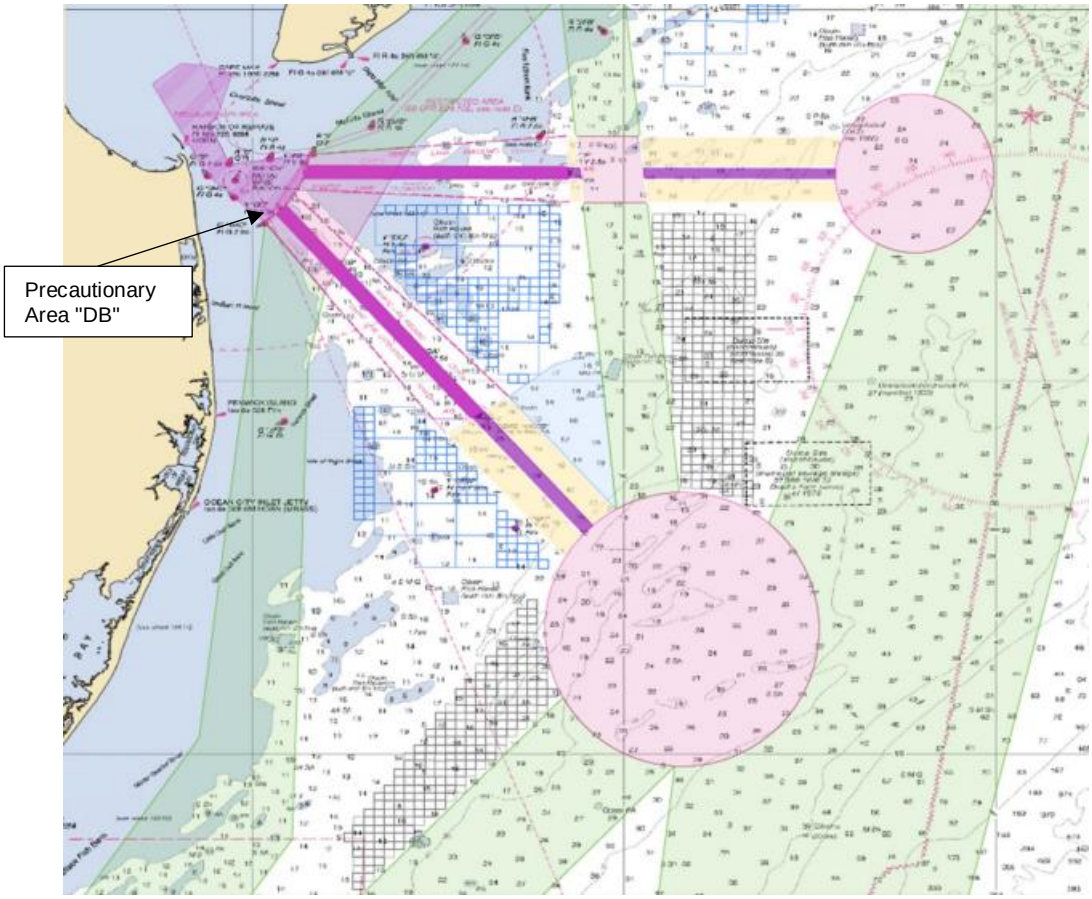

Figure 2: Proposed changes to the TSS Off Delaware Bay with U.S. fairways and offshore energy areas

7 This document proposes an expansion of the precautionary area at the entrance to the Delaware Bay and renames it "DB". This proposed expansion would extend the precautionary area approximately 4.5 NM offshore and would gradually widen to 11 NM, where it would encompass the intersection of traditional shipping routes, and both the Off Delaware Bay: eastern and south-eastern approaches. A precautionary area would signify to mariners that they are transiting through an area, "where ships must navigate with particular caution," owing to the perpendicular crossing of vessel traffic. Figure 2 shows the amendment to the existing precautionary area.

Eastern approach

8 This document proposes to establish a new precautionary area "A" at the offshore terminus of the existing Off Delaware Bay: eastern approach. This proposed precautionary area would be an approximately 29-square nautical mile area, encompassing the intersection of the Off Delaware Bay: eastern approach and traditional shipping routes. The United States expects to see vessel traffic cross perpendicularly to each other at the intersection of the traditional shipping routes and TSS. A precautionary area would signify to mariners that they are transiting through an area, "where ships must navigate with particular caution," because of the perpendicular crossing of vessel traffic.

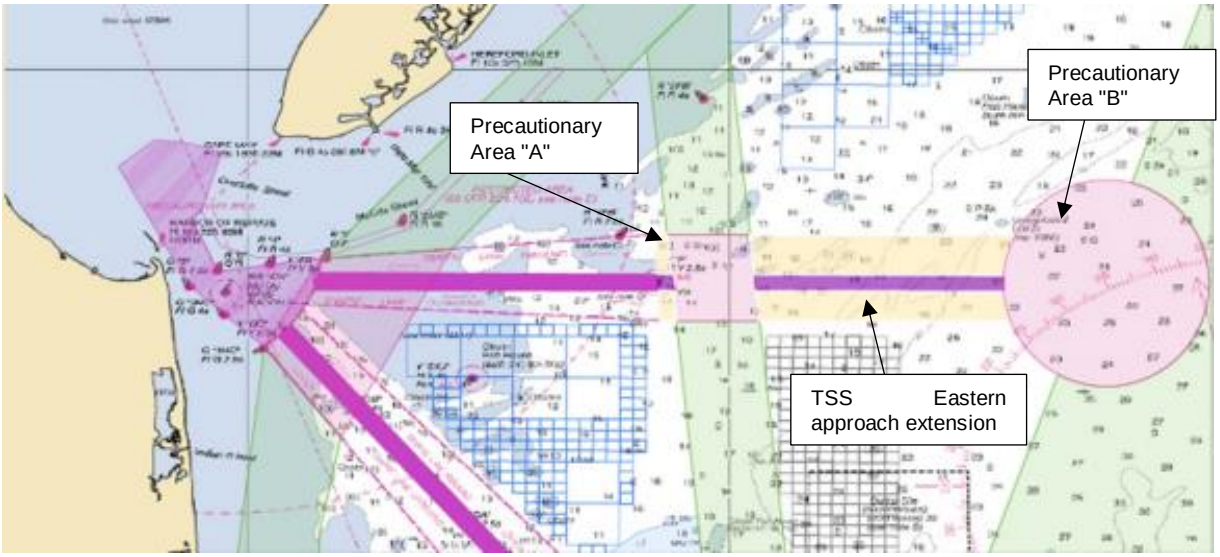

Figure 3: Amendments to the Eastern approach

9 This document proposes an extension of the TSS *Off Delaware Bay*: eastern approach which would extend the TSS separation zone and traffic lanes approximately 16 NM offshore past the proposed precautionary area "A", to where it would intersect with the new precautionary area "B". The United States expects to see considerable growth in the amount of vessel traffic converging at the intersection of the TSS extension and traditional shipping routes described in paragraph 13, and therefore proposes a precautionary area "B" at the intersection. A precautionary area would indicate to mariners that they are transiting through an area "where ships must navigate with particular caution," because of the perpendicular crossing of vessel traffic from the northbound/southbound United States fairway with the TSS. Because the proposed precautionary area "A" would bisect the proposed eastern approach, the proposed separation zone and traffic lanes are presented in two parts in the annex.

South-eastern approach

10 This document proposes an extension of the Off Delaware Bay: south-eastern approach which would extend the TSS separation zone and traffic lanes approximately 12 NM farther offshore and would maintain the width of approximately 5 NM.

11 This document proposes a new precautionary area "C" at the offshore terminus of the proposed extension Off Delaware Bay: south-eastern approach. This proposed precautionary area would be an approximately 314-square nautical mile area encompassing the intersection of the Off Delaware Bay: south-eastern approach and traditional shipping routes. The United States expects to see a considerable amount of vessel traffic meet at the intersection of the traditional shipping routes and TSS. A precautionary area would signify to mariners that they are transiting through an area, "where ships must navigate with particular caution," owing to the perpendicular crossing of vessel traffic.

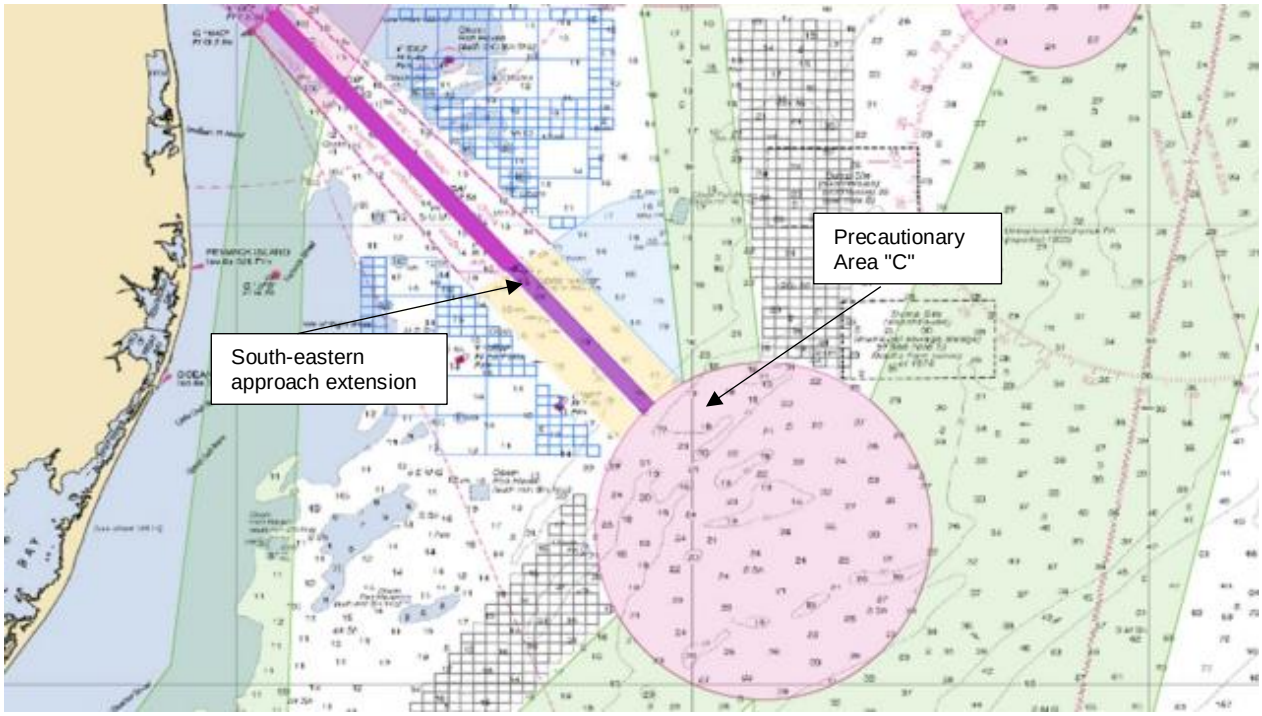

Figure 4: Amendments to the South-eastern approach

Cooperation between interested States

12 The United States has all rights, jurisdiction and obligations to the waters in the proposal to amend the TSS "Off Delaware Bay". The TSS and proposed precautionary areas extend approximately 30 NM offshore and are entirely in the EEZ of the United States, and there is no need for a cooperation agreement with other States.

Traffic considerations

13 The United States analysed vessel traffic off Delaware Bay and focused on waters located seaward of existing port approaches within the United States EEZ. The study referenced in document NCSR 13/INF.6 allowed the United States to consider vessel movements among both domestic and international ports of call to inform marine planning in the area off Delaware Bay. Information used included automatic identification system (AIS) data and information from shipping organizations to identify traditional navigation routes.

The AIS data identified customary routes transited by towing vessels and deep-draught vessels. Because these two vessel types have different manoeuvring capabilities and navigation safety requirements, the identification of their customary routes and required shipping route widths was studied separately and aggregated for the final proposal.

14 Although the most common vessel traffic in the area is entering/departing Delaware Bay through the existing TSS as seen in figure 5. Figure 6 shows consolidated AIS tracks for the predominant northerly/southerly shipping routes. Note a significant amount of vessel traffic that historically transits perpendicular through the existing and proposed TSS and through the offshore energy construction sites seen in black. By establishing the United States fairways as described in paragraph 3, the United States, in consultation with the commercial shipping industry, expect to see a consolidation of shipping traffic into the defined United States fairways. By consolidating traffic into the United States fairways along the Atlantic coast, United States coastwise shipping transits will converge with the TSS in the proposed precautionary areas.

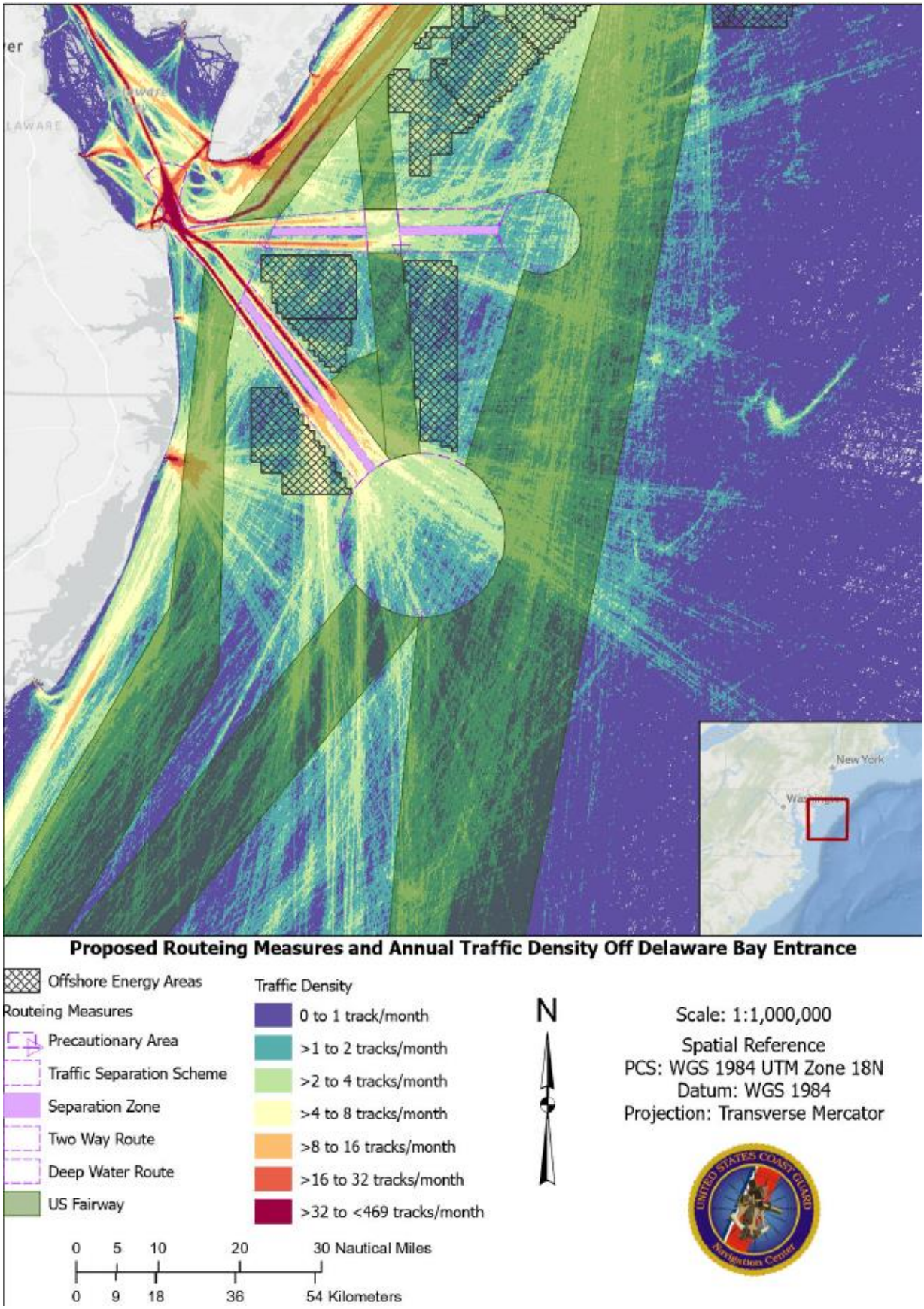

Figure 5: AIS vessel movements off Delaware Bay

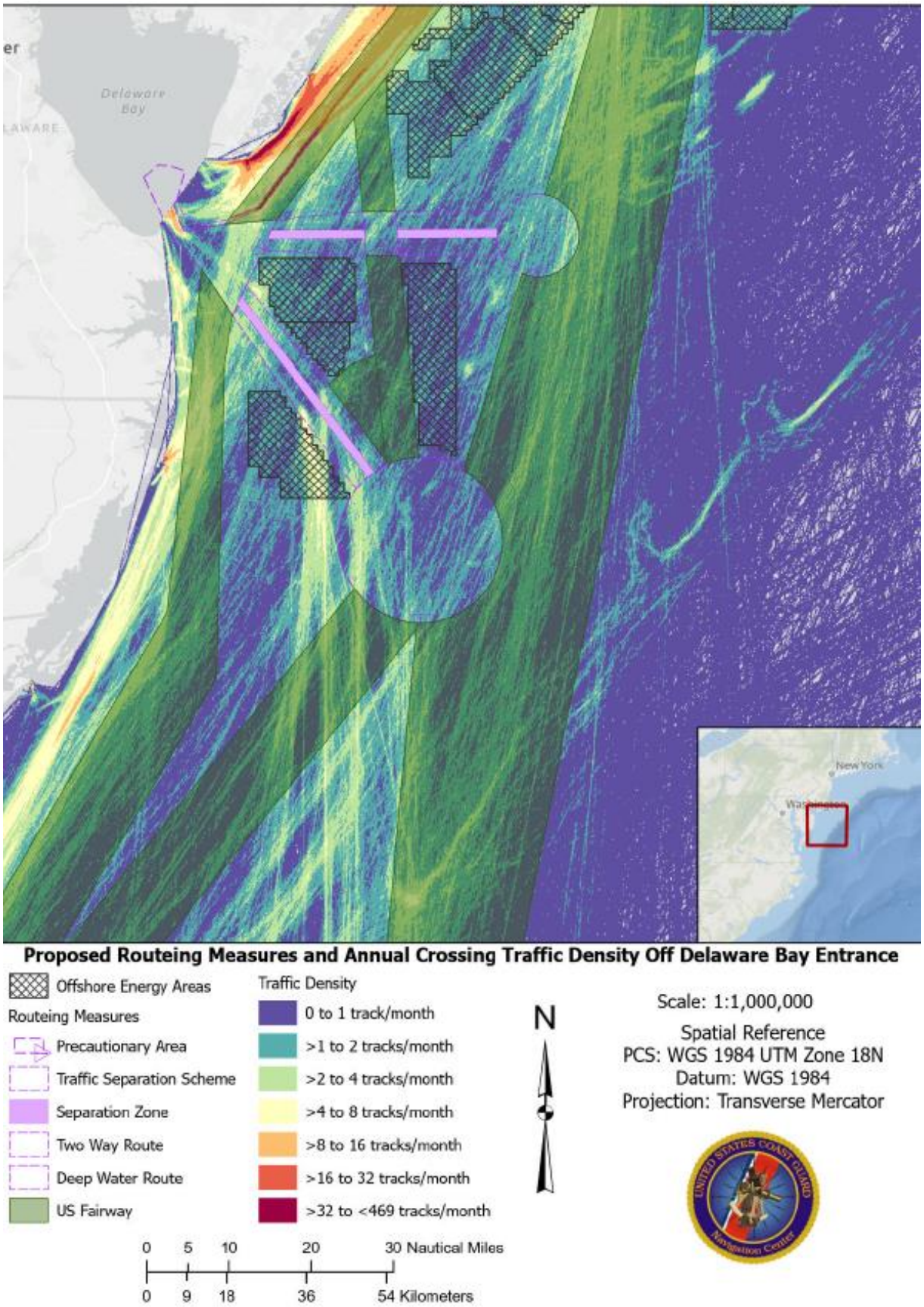

Figure 6: Northerly/southerly AIS vessel movements off Delaware Bay

15 The proposals to extend the eastern and south-eastern TSS are precipitated by offshore construction sites seen in black in figures 5 and 6. Because these sites are about the port access to Delaware Bay, the United States determined the most prudent way to maintain the safety of navigation in the area was to extend the traffic lanes and separation zone. Construction of structures inside the sites will be at least 2 NM apart from the traffic lanes.

16 The existing aids to navigation (AtoN) can be seen in figure 7 and are described in paragraphs 17 through 22.

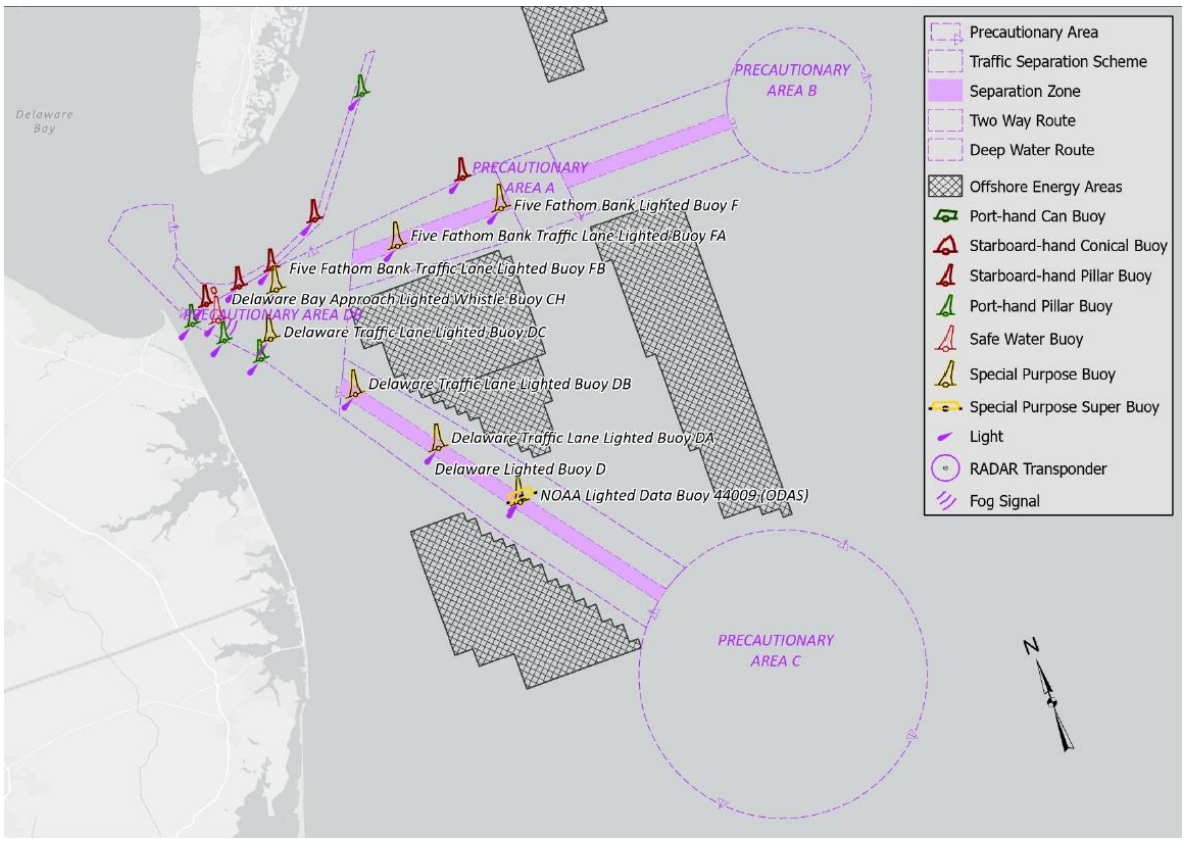

Figure 7: Off Delaware Bay aids to navigation

Eastern approach

17 The eastern approach is marked by three special (yellow) buoys in the centre of the separation zone. These buoys are approximately 7.5 NM apart, labelled, from offshore to inshore, "F", "FA", and "FB". Each buoy is lighted with a distinct yellow flash characteristic of 2.5 seconds ("F"), 4 seconds ("FA"), and 6 seconds ("FB"). The current position of "FB" would place the buoy in the proposed expanded Entrance to Delaware Bay Precautionary Area. A relocation of these AtoN is under consideration to coincide with the adoption of this proposal. The depth of water in the proposed extension of the Eastern approach is sufficient to support physical AtoN.

18 Three red lateral buoys mark the northern boundary of the existing Entrance to Delaware Bay Precautionary Area. These buoys are approximately 2.5 NM apart labelled, from offshore to inshore, "2", "4", and "6". Each buoy is lighted with a red flash characteristic of 1 second ("2"), 4 seconds ("4"), and 1 second ("6"). McCrie Shoal lighted buoy "2MS" is located 4.5 NM to the northeast of buoy "2" and is lighted with a red flash characteristic of 4 seconds. Five Fathom Band NW lighted buoy "3FB" is located 9 NM to the northeast of buoy "2MS" and is lighted with a green flash characteristic of 4 seconds. The United States

does not intend to relocate these buoys upon adoption of this proposal, as they support the two-way route to the north of the precautionary area and do not conflict with the proposed amendments.

19 Two red lateral buoys, "4FB" and "2FB", mark Five Fathom Bank to the north of the Eastern approach. Buoy "4FB" is on the northern border of the inbound traffic lane and is lighted with a red flash characteristic of 2.5 seconds. Buoy "2FB" is located 9.25 NM to the northeast of buoy "4FB" and is lighted with a red flash characteristic of 4 seconds. The United States does not intend to relocate these buoys upon adoption of this proposal, as they mark a shoal to the north of the TSS and do not conflict with the proposed amendments.

South-eastern approach

20 The south-eastern approach is marked by four special (yellow) buoys in the centre of the separation zone. These buoys are approximately 7 NM apart, labelled, from offshore to inshore, "D", "DA", "DB", and "DC". Each buoy is lighted with a distinct yellow flash characteristic of 6 seconds ("D"), 2.5 seconds ("DA"), 4 seconds ("DB"), and 2.5 seconds ("DC"). The current position of "DC" would place the buoy in the proposed expanded Entrance to Delaware Bay Precautionary Area. A relocation of these AtoN is under consideration to coincide with the adoption of this proposal. The depth of water in the proposed extension of the Eastern approach is sufficient to support physical AtoN.

21 Three green lateral buoys mark the southern boundary of the existing Entrance to Delaware Bay Precautionary Area. These buoys are approximately 2.5 NM apart, labelled, from offshore to inshore, "1HC", "3HC", and "5". Each buoy is lighted with a green flash characteristic of 2.5 seconds ("1HC"), 4 seconds ("3HC"), and 2.5 seconds ("5"). These buoys are likely not to be relocated upon adoption of this proposal, as these AtoN also support a separation of near-shore small vessel traffic to the south of the precautionary area and larger vessels using the TSS and precautionary area. These AtoN do not conflict with the proposed amendments.

Entrance to Delaware Bay Precautionary Area "DB"

22 Two AtoN are provided inside the existing Entrance to Delaware Bay Precautionary Area. The centre of the precautionary area is marked by a starboard-hand, preferred channel buoy labeled "DF" with a red, two plus one flash characteristic of 6 seconds. A safe water buoy labelled "CH" with a white morse "A" flash characteristic of 8 seconds is positioned 2 NM to the west of the "DF" buoy. The "CH" buoy also provides a morse "A" RACON and marks a pilot boarding area. These AtoN do not conflict with the proposed amendments and the United States does not intend to relocate upon adoption of this proposal.

23 A virtual AtoN in the proposed areas was considered, but the current AIS AtoN infrastructure does not support the ability to broadcast the virtual AtoN in the proper location at this time. Further consideration of AIS AtoN will be considered in the future as the capability improves.

24 There is no vessel traffic service in the area.

25 The United States National Oceanic and Atmospheric Administration (NOAA) hydrographic surveys for US3DE01M, 2025 edition and US4DE11M, 2025 edition, adhere to International Hydrographic Organization (IHO) standards, including special publication 44. These surveys accurately represent the water depths in the area, and the depths provided on the nautical charts were updated in January 2025. All paper charts in the United States were discontinued by January 2025. Corresponding ENCs are provided for reference and updated in the annex as appropriate.

26 Future offshore energy development is expected along the United States central Atlantic coast. The United States Bureau of Ocean Energy Management (BOEM) has published a proposed "Call Area," which consists of 13,476,805 acres (54,539 square kilometers) off the United States central Atlantic coast (figure 6). While the leased construction sites displayed in figures 1 through 5 show already agreed upon areas for offshore energy production, additional sites could be leased inside the Call Area. Adoption of this proposal, along with the establishment of United States fairways, as discussed in paragraph 3, will prohibit construction in shipping lanes and ensure safe passage to and from the United States ports.

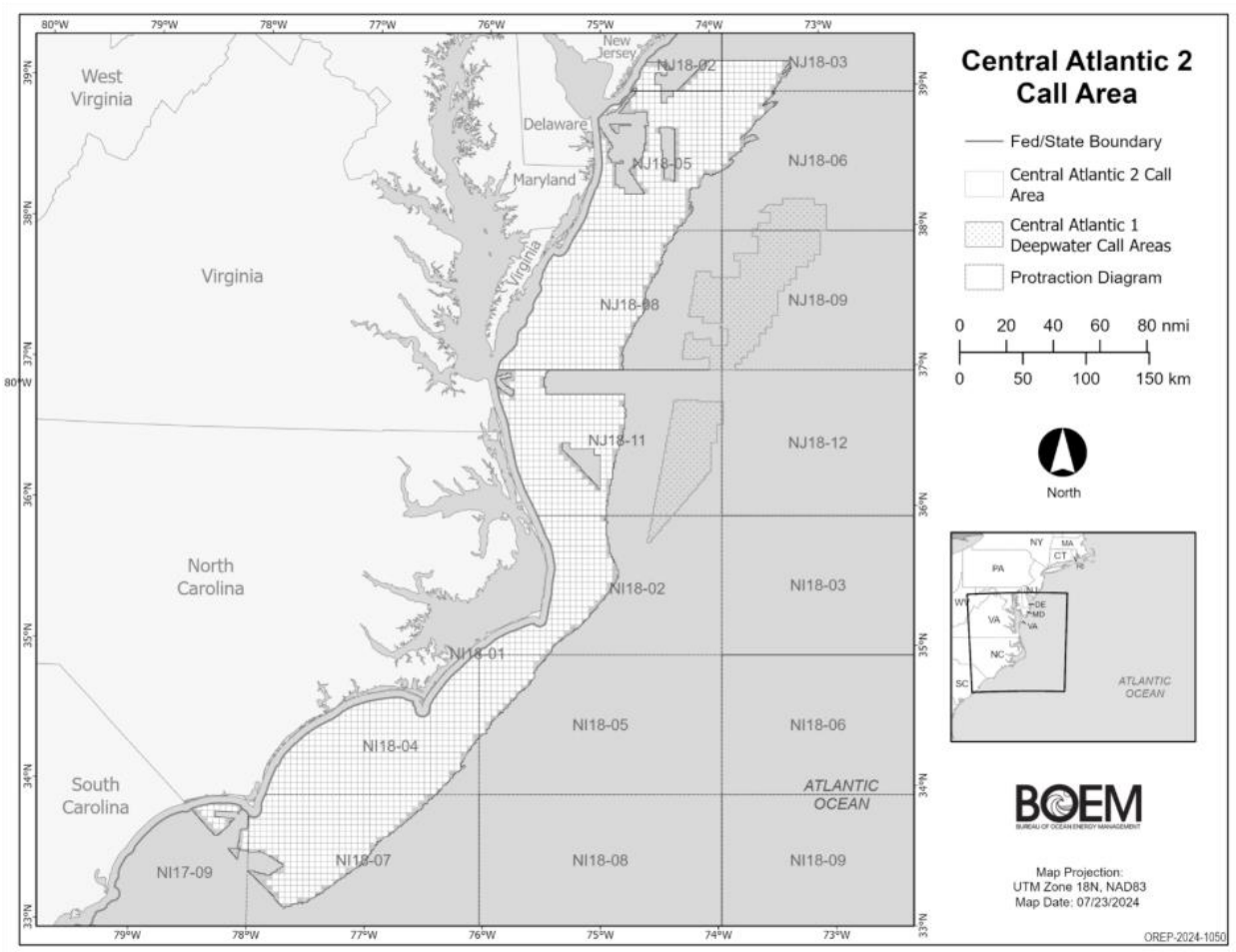

Figure 8: Proposed U.S. offshore energy areas

Marine environmental considerations

27 There are diurnal tides with high and low waters occurring at about the same time off Delaware Bay. The currents have considerable velocity in the inlets along the seacoast and in the narrow channels connecting the inlets and the bays and sounds. Historical data from offshore weather buoys indicate a predominant current to flow in an easterly direction. Seasonal heavy weather occurs in the late summer and mid-winter with high winds of 28 knots or more from the northwest through north from December through March. Hurricane season runs from 1 June through 30 November, with the highest threat of tropical heavy weather in September and October. The Delaware Bay entrance closed for heavy weather three times between 2012 and 2020.

28 The United States considered vessel noise, vessel operations, and vessel movements in the area of the proposal. These considerations are not expected to result in significant impacts to the following resources: air quality, ambient sound, marine vegetation, marine invertebrates, flying insects, birds, bats, marine fish, essential fish habitat, commercial fishing, marine construction, mineral extraction, oil and gas extraction, recreation and tourism, existing renewable energy projects, research, transportation and shipping, and subsistence fishing and hunting.

29 There are no marine environmental considerations that will impact this proposal.

30 Additional information regarding marine and environmental considerations can be found in document NCSR 13/INF.6.

Status of the routeing system

31 This proposal is recommendatory and is designed to preserve traditional maritime commerce routes and safe access to the United States ports and protect them from fixed structures that could impact navigation safety and generally prioritize the right of navigation over all other uses.

Position-fixing in relation to the routeing system

32 Accurate position-fixing is widely available to all vessel traffic using global positioning system (GPS). Aids to Navigation, including the establishment of virtual AtoN, are addressed in paragraphs 16 through 23. Because of low user numbers and significant operational costs compared to satellite-based GPS, LORAN-C was considered unnecessary for modern navigation and discontinued in February 2010. Nautical charts for the area are available. GPS and nautical charts used in conjunction with AIS provide real-time location data to other vessels and shore stations.

Miscellaneous information

33 Prior to establishing the fairways, as discussed in paragraph 3, the United States conducted a Port Access Route study (PARS) to determine traffic density; if current vessel routeing measures are adequate or require modifications; the type of new vessel routeing measures as necessary; and if the usage of the vessel routeing measures should be mandatory for specific classes of vessels. As provided in paragraph 5, a copy of this study is provided in document NCSR 13/INF.6.

34 Mariner participation in a PARS is essential for the United States to gather firsthand operational data, ensuring that proposed, updated, or new fairways and routeing measures safely accommodate vessel traffic while mitigating risks from offshore development and congestion. Delaware Bay pilots, local port authorities, commercial fishing, passenger vessel operators, bulk and container vessel operators, and recreational boaters were consulted throughout the study to capture direct input regarding the proposed area. Pertinent information has been provided in this proposal. Additional information captured from the consultations and subsequent analysis is provided in document NCSR 13/INF.6.

Proposed date of implementation

35 It is proposed that the amendment enter into force six months after adoption by the Committee.

Action requested of the Sub-Committee

36 The Sub-Committee is invited to consider the proposal in the annex and recommend it to the Committee for adoption.

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ANNEX

PROPOSED AMENDED TRAFFIC SEPARATION SCHEME AND ASSOCIATED MEASURES "OFF DELAWARE BAY"

(base document COLREG.2/Circ.42,
as corrected by COLREG.2/Circ.42/Corr.1 and COLREG.2/Circ.42/Corr.2)

The highlighted areas are the proposed changes. Owing to the complexity of the proposed changes, all geographical coordinates have been renumbered except for the two-way Route off Delaware Bay.

OFF DELAWARE BAY

(Reference charts: United States ~~12214, 1994 edition~~ US3DE01M, 2025 edition; US4DE11M, 2025 edition.

Note: ~~This~~These charts are based on ~~North American datum 1983~~World Geodetic System of 1984 datum.)

Description of the traffic separation scheme

The traffic separation scheme "Off Delaware Bay" consists of three parts:

Part I

Precautionary Areas

The Entrance to Delaware Bay Precautionary Area "DB" is established as follows:
from (29) 38°42′.80 N, 074°58′.90 W;
then south-easterly to (26) 38°37′.25 N, 074°54′.15 W;
then north-easterly to (9) 38°48′.89 N, 074°47′.29 W;
then westerly to (30) 38°48′.32 N, 074°55′.30 W;
then westerly to (31) 38°47′.50 N, 075°01′.80 W;
then northerly to (32) 38°50′.75 N, 075°03′.40 W;
then north-easterly to (33) 38°51′.27 N, 075°02′.83 W;
then northerly to (34) 38°54′.80 N, 075°01′.60 W;
then westerly by an arc of 6.7 nautical miles centred at (38) 38°48′.83 N, 075°05′.51 W to (35) 38°55′.53 N, 075°05′.87 W;
then south-westerly to (36) 38°54′.00 N, 075°08′.00 W;
then southerly to (37) 38°46′.60 N, 075°03′.55 W;
then south-easterly to (29) 38°42′.80 N, 074°58′.90 W.

Precautionary Area "A" off Delaware Bay:
Eastern Approach is an area bounded by a line connecting the following geographical points:

(10) 38°49′.80 N, 074°33′.91 W (14) 38°44′.45 N, 074°33′.32 W
(11) 38°49′.79 N, 074°28′.74 W (15) 38°44′.44 N, 074°28′.15 W

Precautionary Area "B" off Delaware Bay:
Eastern Approach is an area with a radius of 5 NM centered upon geographical position (39) 38°46′.79 N, 074°06′.60 W.

Precautionary Area "C" Off Delaware Bay:

Southeastern Approach is an area with a radius 10 NM centered upon geographical position (40) 38°09′.88 N, 074°25′.20 W.

Part II

Eastern approach

(a) A separation zone is established bounded by a line connecting the following geographic positions:

(1) 38°46′.60 N,	074°12′.99 W	(3) 38°47′.62 N,	074°28′.50 W
(2) 38°46′.61 N,	074°28′.39 W	(4) 38°47′.60 N,	074°12′.91 W

(~~a~~b) A separation zone is established bounded by a line connecting the following geographic positions:

(~~1~~5) 38°46′.~~30~~61 N,	074°~~34′.45~~33′.56 W	(~~3~~7) 38°47′.~~45~~59 N,	074°~~55′.40~~48′.06 W
(~~2~~6) 38°46′.~~33~~58 N,	074°~~55′.75~~48′.65 W	(~~4~~8) 38°47′.~~35~~61 N,	074°~~34′.50~~33′.67 W

(~~b~~c) A traffic lane for westbound traffic is established between the separation zone and a line connecting the following geographic positions:

(~~5~~9) 38°48′.~~32~~89 N	074°~~55′.30~~47′.29 W	(~~6~~10) 38°49′.80 N,	074°~~34′.60~~33′.91 W

(d) A traffic lane for westbound traffic is established between the separation zone and a line connecting the following geographic positions:

(11) 38°49′.79 N, 074°28′.74 W	(12) 38°49′.77 N, 074°11′.74 W

(~~c~~e) A traffic lane for eastbound traffic is established between the separation zone and a line connecting the following geographic positions:

(~~7~~13) 38°45′.~~45~~14 N,	074°~~56′.20~~49′.50 W	(~~8~~14) 38°44′.45 N,	074°~~34′.35~~33′.32 W

(f) A traffic lane for eastbound traffic is established between the separation zone and a line connecting the following geographic positions:

(15) 38°44′.44 N, 074°28′.15 W	(16) 38°44′.43 N, 074°12′.24 W

Part ~~II~~ III

South-eastern approach

(a) A separation zone is bounded by a line connecting the following geographic positions:

(17) 38°17′.59 N, 074°33′.31 W	(~~11~~20) 38°~~44′.20~~39′.58 N, 074°~~57′.20~~52′.78 W
(~~9~~18) 38°27′.00 N, 074°42′.30 W	(~~12~~21) 38°27′.60 N, 074°41′.30 W
(~~10~~19) 38°~~43′.40~~38′.58 N, 074°~~58′.00~~53′.37 W	(22) 38°18′.18 N, 074°32′.31 W

(b) A traffic lane for north-westbound traffic is established between the separation zone and a line connecting the following geographic positions:

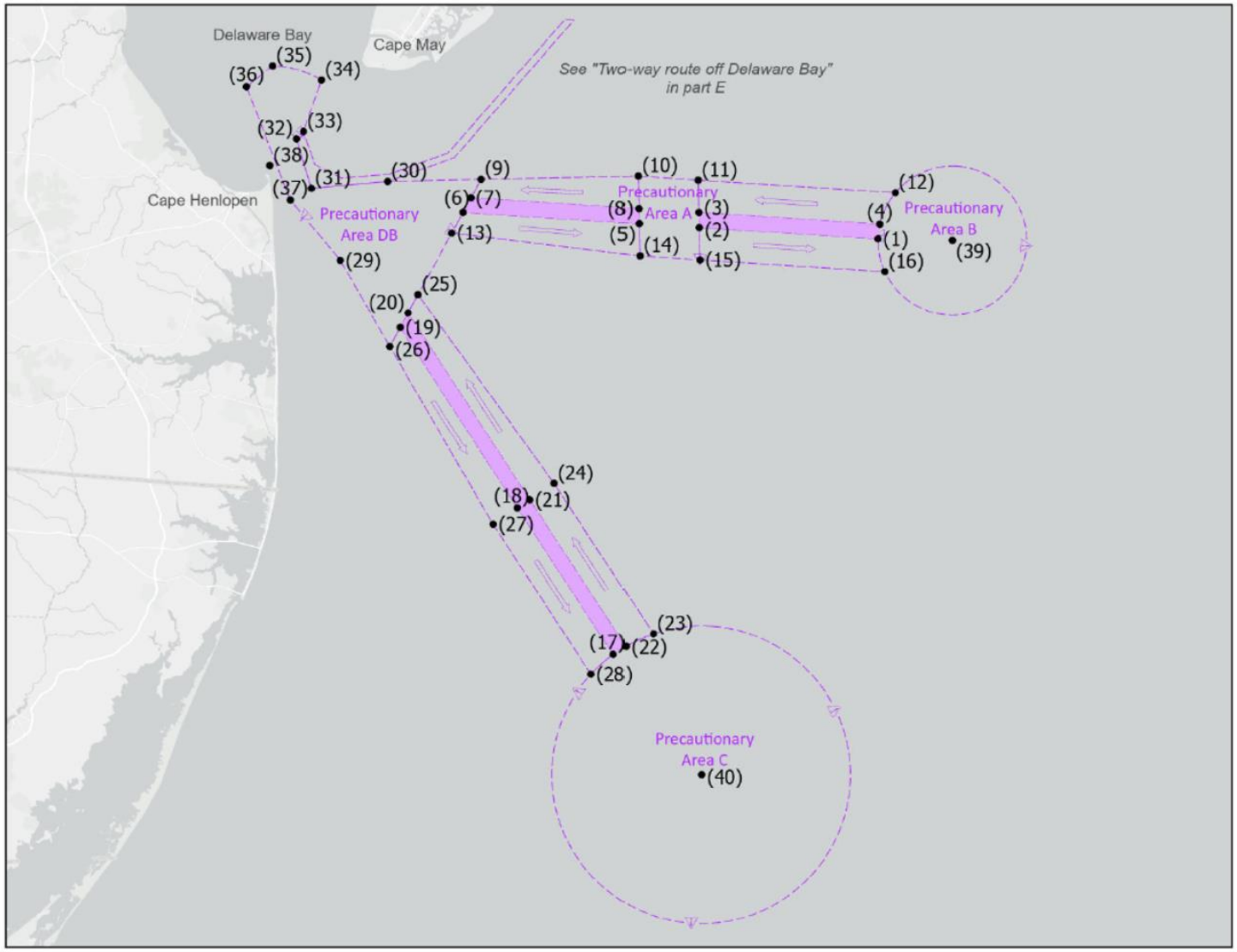
(23) 38°19′.13 N, 074°30′.07 W
(~~13~~24) 38°28′.80 N, 074°39′.30 W
(~~14~~25) 38°~~45′.10~~40′.85 N, 074°~~56′.60~~52′.03 W

(c) A traffic lane for south-eastbound traffic is established between the separation zone and a line connecting the following geographic positions:

(~~15~~26) 38°~~42′.80~~37′.25 N, 074°~~58′.90~~54′.15 W
(~~16~~27) 38°~~27′.00~~25′.78 N, 074°~~45′.40~~44′.28 W
(28) 38°16′.16 N, 074°35′.09 W

~~**Precautionary area**~~

~~A precautionary area is established as follows: from 38°42.80′ N, 074°58.90′ W; then northerly by an arc of 8 nautical miles centred at 38°48.90′ N, 075°05.60′ W to 38°48.32′ N, 074°55.30′ W; thence westerly to 38°47.50′ N, 075°01.80′ W; thence northerly to 38°50.75′ N, 075°03.40′ W; thence north-easterly to 38°51.27′ N, 075°02.83′ W; thence northerly to 38°54.80′ N, 075°01.60′ W; thence westerly by an arc of radius 6.7 nautical miles centred at 38°48.90′ N, 075°05.60′ W to 38°55.53′ N, 075°05.87′ W; thence south-westerly to 38°54.00′ N, 075°08.00′ W; thence southerly to 38°46.60′ N, 075°03.55′ W; then south-easterly to 38°42.80′ N, 074°58.90′ W.~~


Delaware Bay
Cape May
See "Two-way route off Delaware Bay" in part E
Cape Henlopen
Precautionary Area DB
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Precautionary Area B
Precautionary Area C
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